

SKM4950B Presentation Semester 1 Session 2025/2026

ToothyMate: Smart Oral Health Awareness with AR & AI

in collaboration with
Klinik Pergigian Dr Karthi

Presented by:

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INTRODUCTION TO TOOTHYMATE



The Urgency: Pediatric Oral Health Crisis

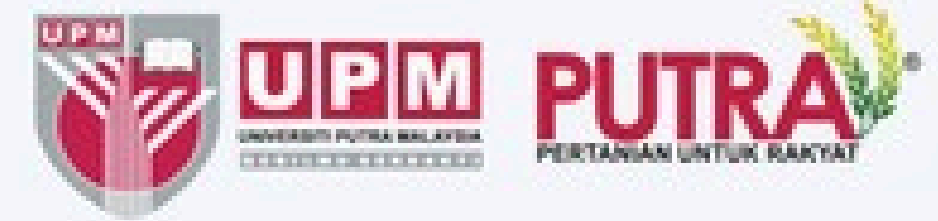
- Global Scale: Dental caries affects 514 million children worldwide, making it the #1 chronic pediatric disease (WHO, 2022).
- Local Reality: 63.4% of Malaysian preschoolers have untreated caries (Mohd Nor et al., 2020); prevalence reaches 61.6% among indigenous children (Abdul Kadir et al., 2019).



The Limitations of Current Approaches

- Traditional Methods: Classroom instructions and pamphlets are passive, uninteresting, and fail to sustain long-term behavioral change (Romalee et al., 2023).
- Existing Mobile Apps: Current market solutions lack depth, averaging only 6.87/31 gamification features and often missing comprehensive educational content (Schwarz et al., 2020; Tezcan et al., 2025).

INTRODUCTION TO TOOTHYMATE

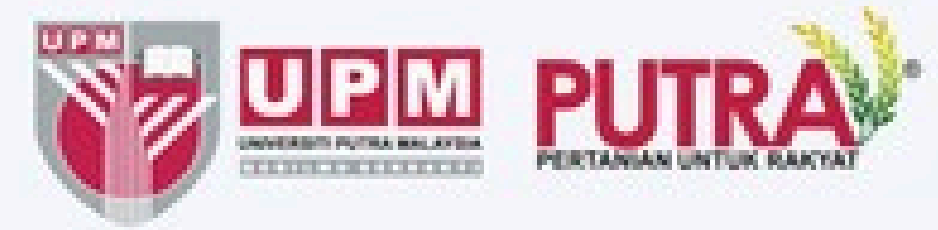


The Solution: ToothyMate



- **Innovation:** A gamified ecosystem integrating Augmented Reality (AR) for visualization and Artificial Intelligence (AI) for feedback.
- **Engagement:** Transforms hygiene into an adventure with Missions, XP, and Hero Ranks.
- **Collaboration:** Developed in partnership with Klinik Pergigian Dr. Karthi.

PROBLEM STATEMENT



1

Limited Engagement with Traditional Oral Health Education for Children

- Traditional methods (pamphlets, lectures) are passive and fail to resonate with young audiences.
- Existing apps lack depth, averaging only 6.87 out of 31 gamification features (Schwarz et al., 2020).
- Current tools are insufficient for sustaining long-term behavioral change (Tezcan et al., 2024).

2

Difficulty in Visualizing Abstract Dental Concepts

- Children struggle to comprehend "invisible" processes like decay through static images.
- AR significantly improves performance-based learning but is underutilized in dental education (Garzón et al., 2022).
- There is a lack of tools designed to make abstract 3D concepts tangible and understandable (Romalee et al., 2023).

3

Lack of AI-Driven Educational Feedback

- Monitoring is limited to infrequent professional visits, leaving parents without home assessment tools.
- Existing AI models focus on clinical diagnosis rather than motivating children (Kühnisch et al., 2022).
- Lack of accessible tools that link detection directly to encouragement and e-learning (Alharbi et al., 2024).

OBJECTIVES



1

To develop a **gamified mobile application** aimed at improving oral health awareness

2

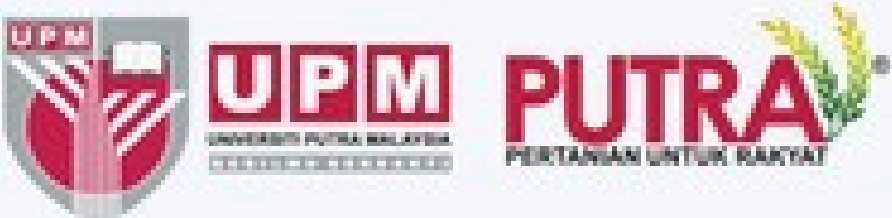
To incorporate **Augmented Reality (AR)** for immersive exploration of 3D tooth anatomy and dental conditions.

3

To develop an **on-device TensorFlow Lite model** that classifies dental conditions for educational feedback.

LITERATURE REVIEW MATRIX

Engagement & Interaction (Gamification & AR)



Title & Citation	Features & Methods	Key Findings	Limitations	How ToothyMate Solves It
The global prevalence of early childhood caries <i>(Uribe et al., 2021)</i>	Systematic review of ECC prevalence	ECC affects ~50% of preschool children globally	Regional variations in diagnostic criteria	Provides accessible early education to parents & kids to reduce ECC prevalence.
The Application of Gamification in Children’s Oral Health <i>(Tezcan et al., 2025)</i>	Review of 41 gamification studies	80% studies showed effectiveness in plaque control	Limited long-term follow-up	Implements reward systems (badges/points) to sustain long-term interest.
The Effects of Gamification and Oral Self-Care <i>(Schwarz et al., 2020)</i>	App store analysis of 17 apps	<20% of apps include comprehensive education	Most apps lack educational content	Uses interactive quizzes to verify understanding of the dental syllabus.
Ten years of augmented reality in education <i>(Garzón et al., 2022)</i>	Meta-analysis of 134 studies	AR benefits learning outcomes & performance	Treatment duration affects outcomes	Uses interactive AR to visualize dental cases on a 3D model, allowing repeated exploration.
A mobile AR-integrated oral health education <i>(Romalee et al., 2023)</i>	MAR app using 3D models	Improved oral knowledge and self-efficacy	Lower usability scores	Offers a user-friendly AR interface designed specifically for children.

LITERATURE REVIEW MATRIX

AI Technology & Clinical Effectiveness



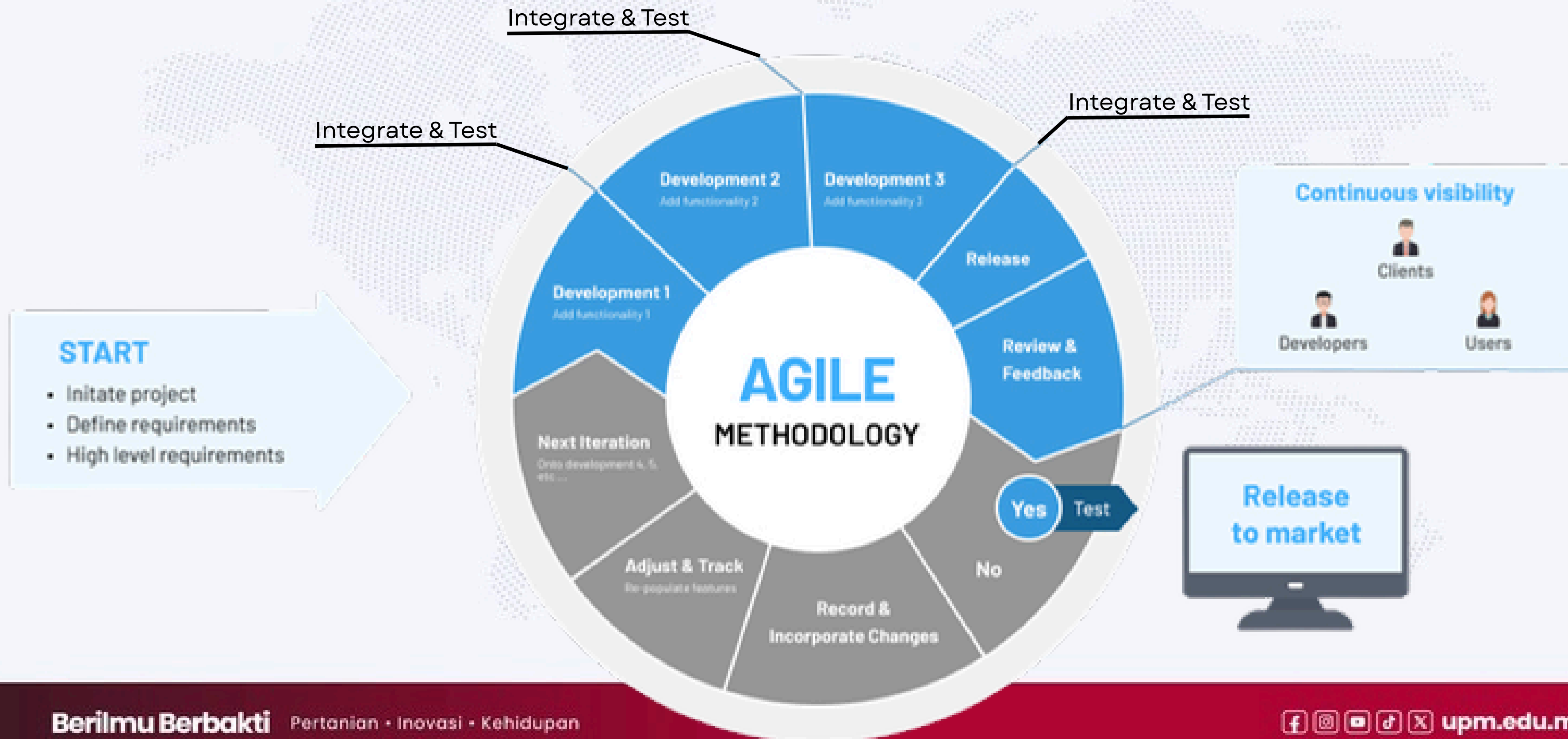
Title & Citation	Features & Methods	Key Findings	Limitations	How ToothyMate Solves It
Caries Detection on Intraoral Images Using Artificial Intelligence <i>(Kühnisch et al., 2022)</i>	MobileNetV2 CNN for caries detection from photos	>90% accuracy, expert-level performance	Ideal images only; excluded other conditions	Uses AI classification to screen and identify multiple conditions (Caries, Plaque, Stain), covering what the study excluded.
Deep Learning Approach to Automatic Tooth Caries Segmentation <i>(Alharbi et al., 2024)</i>	Attention U-Net for caries detection	Sensitivity 0.83, Precision 0.91	Limited to panoramic radiographs	Adopts similar Deep Learning tech to categorize oral health issues automatically.
The use of AI to aid in oral hygiene education: A scoping review <i>(Tay et al., 2023)</i>	Scoping review of 20 AI studies	AI practical for daily use & can combine with AR	Privacy and image quality concerns	Integrates AI Chatbot to explain the classification results (e.g., "What is Plaque?").
Impact of Dent-O-Xpert on oral health knowledge and behavior <i>(Nagarajappa et al., 2023)</i>	RCT comparing app vs traditional	96.4% brushing twice daily post-intervention	Self-reported data accuracy	Digitizes tracking (logs) to encourage consistent twice-daily brushing habits.
Effectiveness of OSCA for Precision Oral Self-Care <i>(Chang et al., 2021)</i>	App based on Behavior Change Wheel	Higher frequency of use = greater improvement	Short-term study duration	Uses 'Streaks' to ensure high-frequency usage.

COMPARISON WITH EXISTING APPS



Application	Gamification	AR	AI Classification	E-Learning	Bilingual (EN/BM)	Child-Focused	Additional Hardware
Brush DJ <i>(Underwood, 2025)</i>	Limited	No	No	No	No	No	No
Oral-B App <i>(Procter & Gamble, 2026)</i>	Yes	No	Yes (brushing)	Limited	No	Limited	Yes
Pokémon Smile <i>(The Pokémon Company, 2025)</i>	Yes	Yes	No	No	No	Yes	No
Colgate Connect <i>(Colgate-Palmolive, 2025)</i>	Yes	No	Limited	Limited	No	Limited	Yes
Dental Monitoring <i>(Dental Monitoring SAS, 2025)</i>	No	No	Yes	No	No	No	Yes (ScanBox)
ToothyMate <i>(This Project)</i>	Yes	Yes	Yes	Yes	Yes	Yes	No

PROJECT METHODOLOGY

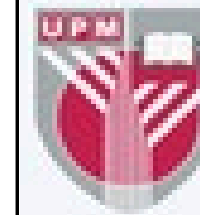
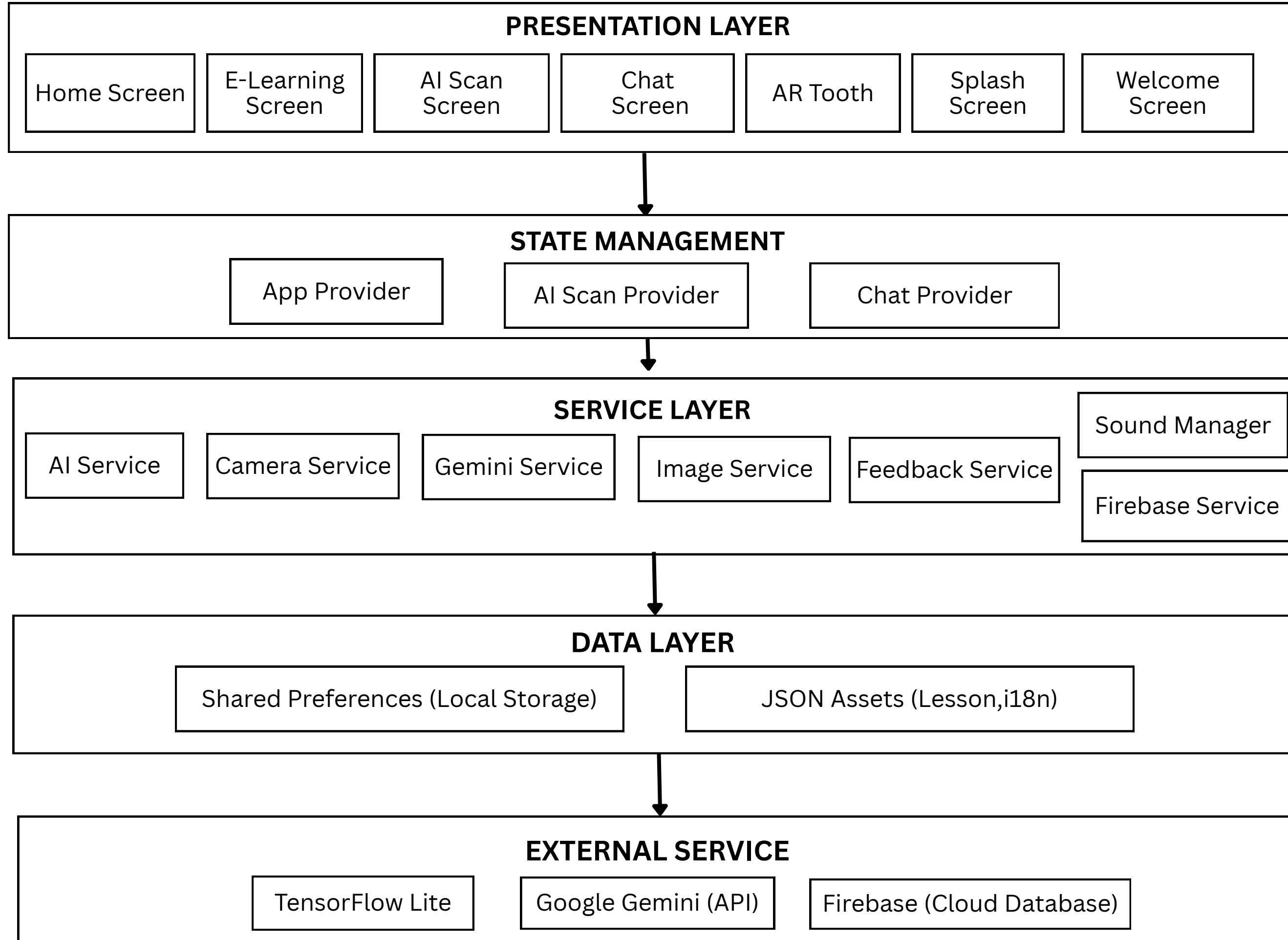




Iteration	Module	Key Features
1	Gamification	XP system, streaks, badges, hero ranks, daily missions
2	E-Learning	13 lessons, flip cards, quizzes, YouTube videos, bilingual (EN/MS)
3	AR Module	3D tooth model, gesture controls, 5 discoverable cases
4	AI Scan	TensorFlow Lite, camera input, 5-class classification
5	AI Chatbot	Google Gemini API, emotion detection, text-to-speech
6	Firebase	Anonymous authentication, Firestore, cloud synchronization
7	Integration & Polish	Localization (EN/MS), sound effects, Lottie animations

AGILE ITERATIVE
INCREMENTAL
DEVELOPMENT

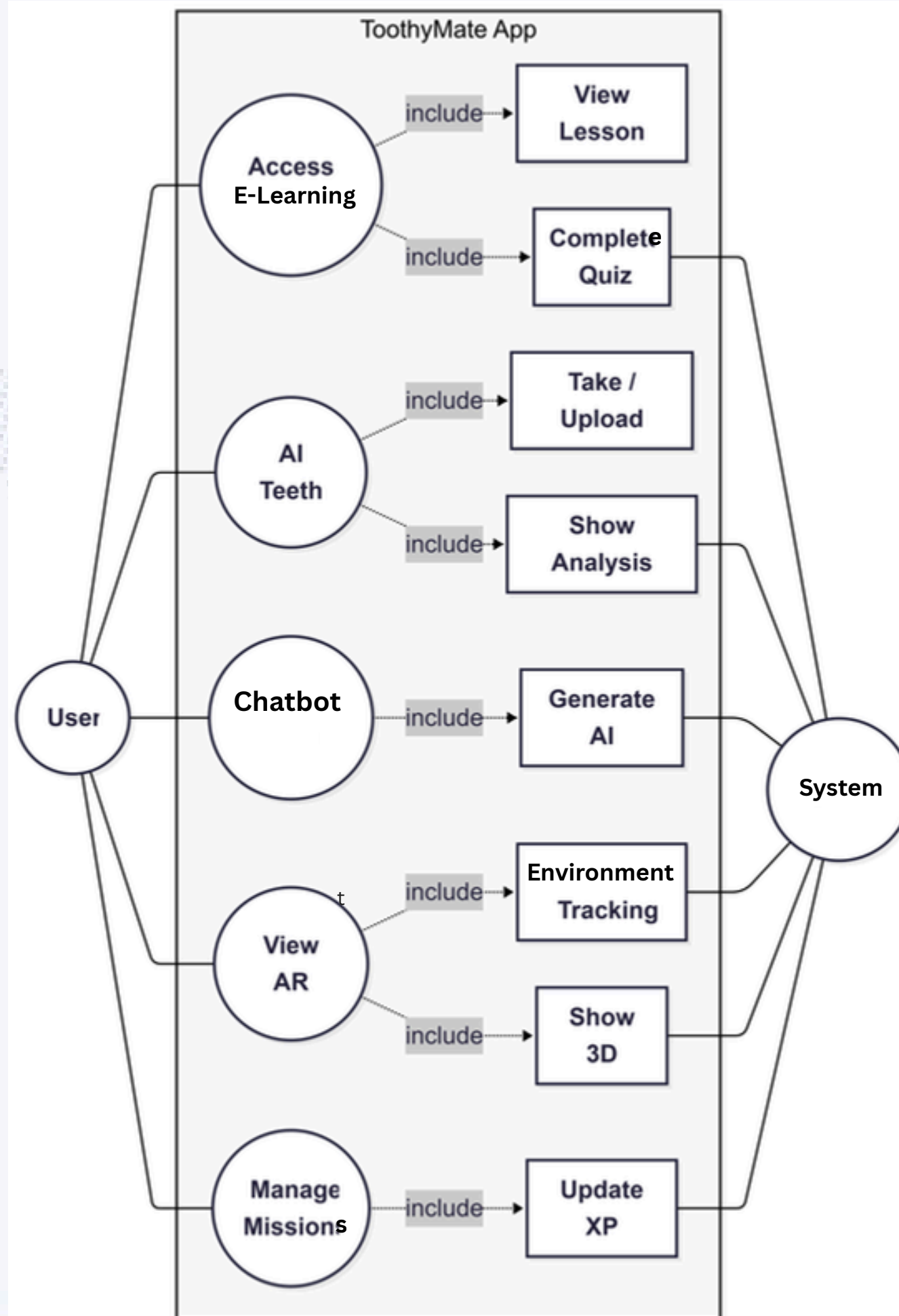
TOOTHYMATE ARCHITECTURE



SYSTEM ARCHITECTURE

Model-View-Provider

SYSTEM USE CASE DIAGRAM

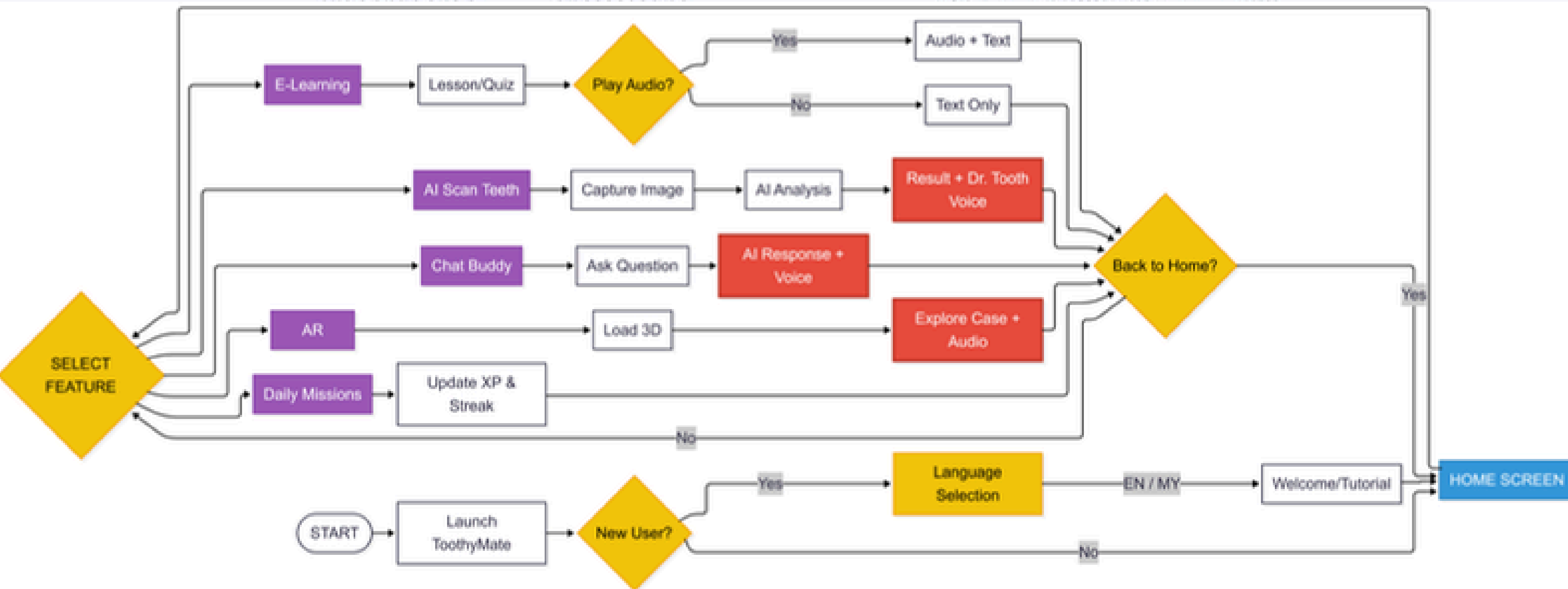


USER FLOWCHART



UPM
UNIVERSITI PUTRA MALAYSIA

PUTRA
PERTANIAN UNTUK RAKYAT



TECHNOLOGY STACK



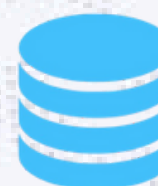
FRONTEND



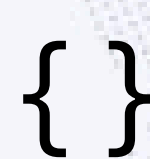
BACKEND



Firebase Firestore
Firebase Anonymous Auth



SharedPreferences



JSON Assets
JSON i18n

ACCESSIBILITY & AUDIO

- Flutter TTS
- Audioplayers

Artificial Intelligence



TensorFlow Lite



Augmented Reality



ARCore



blender



TRAINING DATASET

Amr-ma9e3. (2024). Teeth Dataset. Roboflow Universe.
<https://universe.roboflow.com/rgb-gvcor/caries-novip>

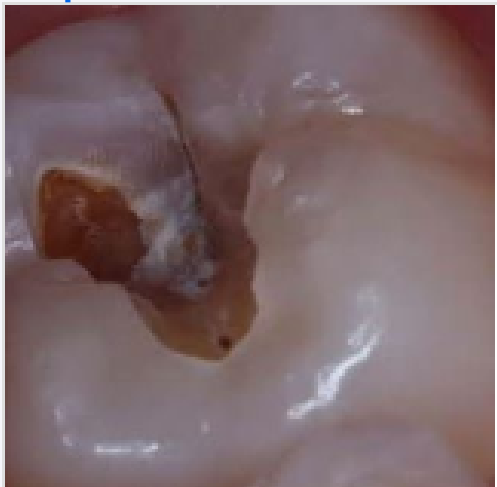


caries Object Detection Model by RGB

5966 open source caries images plus a pre-trained caries model and API. Created by RGB



Zafair. (2024). Carious Classifier Dataset. Roboflow Universe.
<https://universe.roboflow.com/zafair/carious-classifier-zbydx>



Carious Classifier Classification Model by zafair

1122 open source Healthy-vs-Carious images plus a pre-trained Carious Classifier model and API. Created by zafair



teeth Object Detection Model by amr

821 open source teeth images plus a pre-trained teeth model and API. Created by amr



Rgb-gvcor. (2024). Caries Detection Dataset. Roboflow Universe.
<https://universe.roboflow.com/rgb-gvcor/caries-novip>



Healthy Teeth Object Detection Model by Sultan

235 open source Teeth images plus a pre-trained Healthy Teeth model and API. Created by Sultan

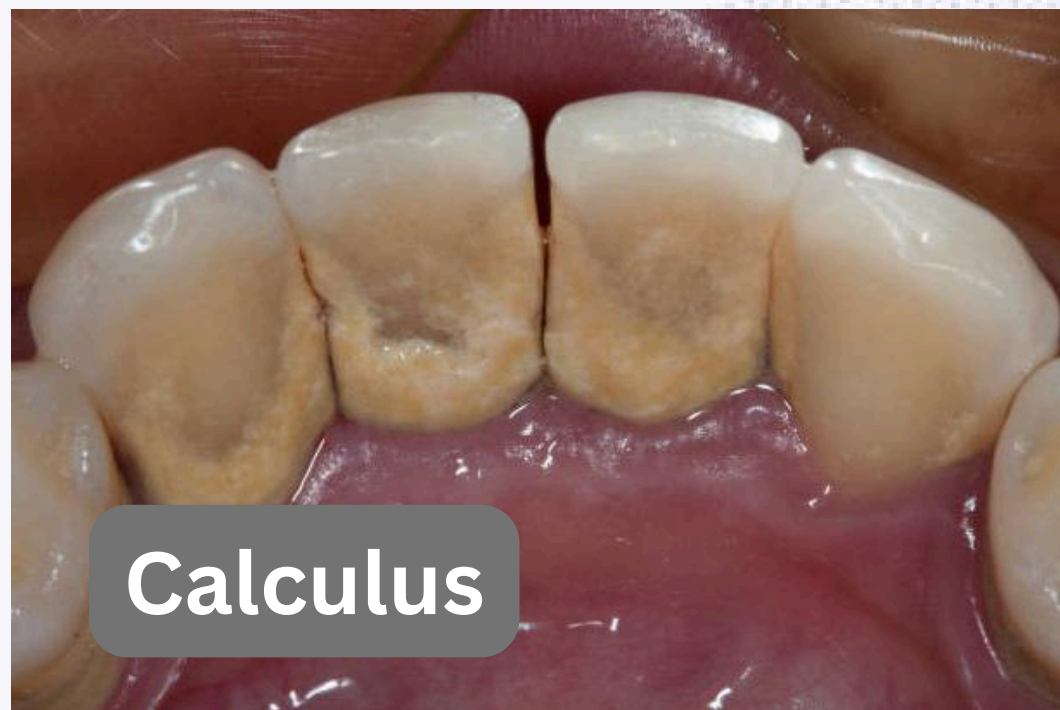


Sultan-qyobm. (2024). Healthy Teeth Dataset (Version 1) [Data set]. Roboflow Universe. <https://universe.roboflow.com/sultan-qyobm/healthy-teeth-hgddf>



Klinik Pergigian Dr. Karthi. (2025). Private Dental Image Dataset for ToothyMate Project Collaboration. [Unpublished raw data].

TEETH CASES CLASSIFICATION



Calculus



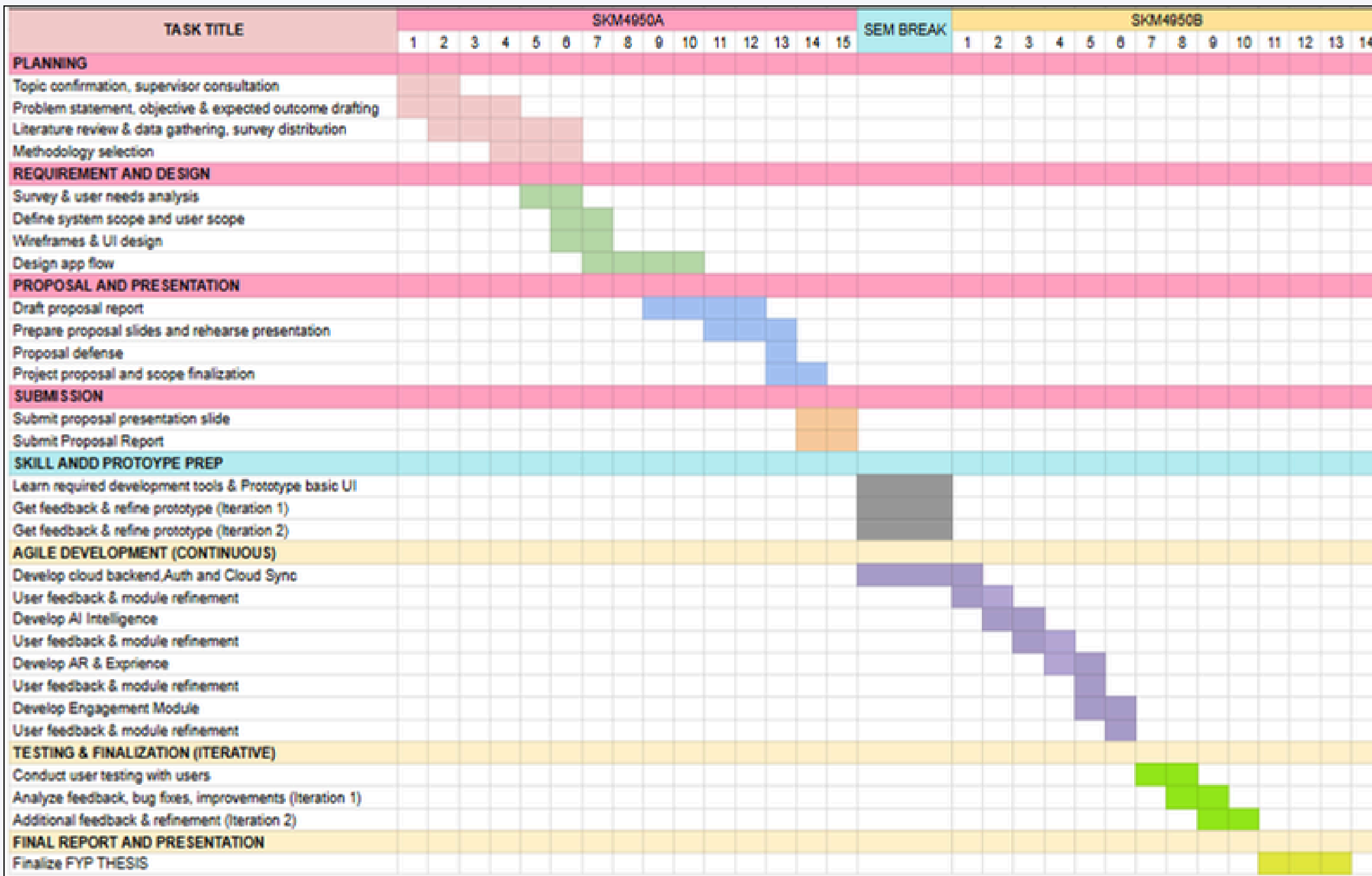
Stain



Healthy Teeth



Caries



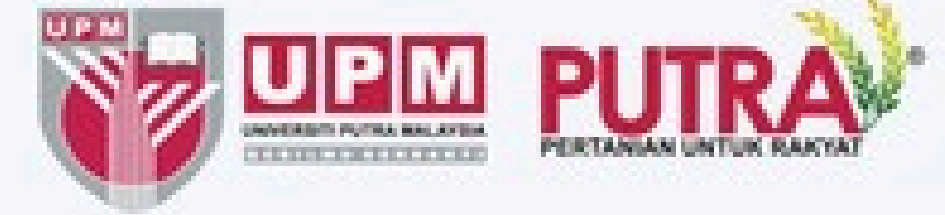






GANTT CHART

REQUIREMENTS



Functional Requirements

User & Data Management

- Anonymous user authentication
- Local progress storage
- Cloud sync (Firestore)
- Offline access

Gamification System

- Daily missions
- XP, levels & streaks
- Badges & hero ranks

AR Learning

- 3D tooth visualization
- Gesture interaction
- 10 interactive cases
- Audio guidance

E-Learning Module

- Lessons & quizzes
- Educational videos
- Progress tracking

AI Features

- Dental image scanning
- Educational feedback
- AI chatbot guidance

Non-Functional Requirements

Performance

- App launch < 3s
- AI classification < 2s
- AR loading < 5s
- Chat response < 3s
- AR frame rate 30+ FPS

Usability

- Child-friendly interface
- Simple navigation
- Text-to-speech support
- English & Malay
- First-time tutorial

Reliability

- Offline caching
- Auto-save progress
- Error handling
- Reliable cloud sync

Security

- Anonymous authentication
- User data privacy rules
- AI safety filtering
- Age-appropriate content

Compatibility

- Android 7.0+ • ARCore devices

1. USER TESTING (USABILITY)

- **Participants:** 21 users
- **Demographics:** Children (15), Parents (4), Dental (2)
- **Method:** Google Forms survey
- **Metrics:** SUS Score, Feature ratings, Satisfaction

2. AI MODEL TESTING (ACCURACY)

- **Test dataset:** Separate validation images
- **Metrics:** Accuracy, Precision, Recall, F1-Score
- **Method:** Confusion matrix analysis
- **Comparison:** Multiple model architectures

USER TESTING RESULTS



Participant Demographics (N = 21)

Category	Count	Percentage (%)
Child Users	15	71.4
Parents / Guardians	4	19
Dental Professional	2	9.5
Total	21	100

Age Distribution (Children)

Age Group (Years)	Count
Under 6	3
7 – 10	4
11 – 12	11
Total	18

USER TESTING RESULTS



Feature Engagement Ratings

Feature Selected	Count	Percentage (%)
Learning lessons & quizzes	21	100
AR tooth visualization	20	95.2
AI-assisted oral hygiene awareness	20	95.2
Rewards / progress tracking	19	90.5
Overall design & animations	19	90.5

Satisfaction Scores (5-Point Likert Scale)

Question	Mean	SD
App helps children become aware of oral hygiene	5	0
App encourages regular brushing	5	0
Activities promote positive daily habits	5	0
Sound effects & quizzes are interactive	5	0
App provides enough dental care information	4.95	0.22
Gamification motivates children	5	0
3D visualization is clear and engaging	5	0
AI feedback is easy to understand	5	0
Language is easy to understand	5	0

SUS SCORE CALCULATION & RESULTS



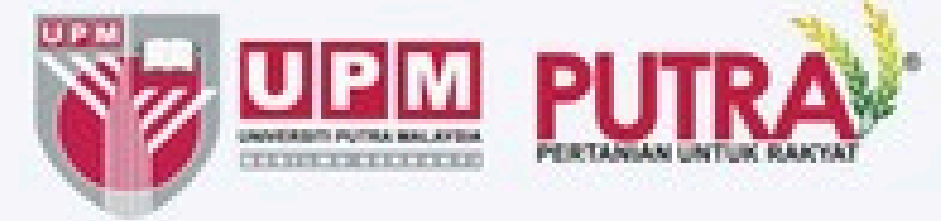
SUS Questions – Mean Responses

	Question Summary	Type	Mean
1	Use app frequently	Positive	5
2	App is complex	Negative	1.1
3	App is easy to use	Positive	5
4	Need technical support	Negative	1.19
5	Functions well integrated	Positive	4.9
6	App is inconsistent	Negative	1
7	Easy to learn	Positive	4.9
8	App is cumbersome	Negative	1
9	Interface is clear	Positive	4.95
10	Need to learn a lot	Negative	1.24

SUS Score Results

Metric	Value
Mean SUS Score	97.38
Minimum Score	85
Maximum Score	100
Standard Deviation	4.21
Grade	A+ (Best Imaginable)
Industry Average	68

AI MODEL ACCURACY TESTING



Class	TP	FP	FN	Precision (%)	Recall (%)	F1-Score (%)
Calculus	16	1	4	94.1	80	86.5
Caries	18	0	2	100	90	94.7
Stain	19	8	1	70.4	95	80.9
Healthy	18	1	2	94.7	90	92.3
Others	19	0	1	100	95	97.4
Average (Macro)	–	–	–	91.8	90	90.4

Metric	Value
Overall Accuracy	90%
Macro Precision	91.80%
Macro Recall	90.00%
Macro F1-Score	90.40%
Total Test Images	100
Images per Class	20

ML MODEL COMPARISON



MobileNetV2 (Fine-Tuned)
Accuracy: 88.24%

True Label	Calculus	Caries	Healthy_Teeth	Stain	not_teeth
	22	0	0	1	0
	0	21	2	0	0
	1	1	19	0	2
	5	2	0	16	0
	0	0	0	0	27
		Predicted Label			

ResNet50
Accuracy: 44.54%

True Label	Calculus	Caries	Healthy_Teeth	Stain	not_teeth
	19	0	0	1	3
	12	6	1	1	3
	9	8	2	3	1
	8	1	0	6	8
	5	0	1	1	20
		Predicted Label			

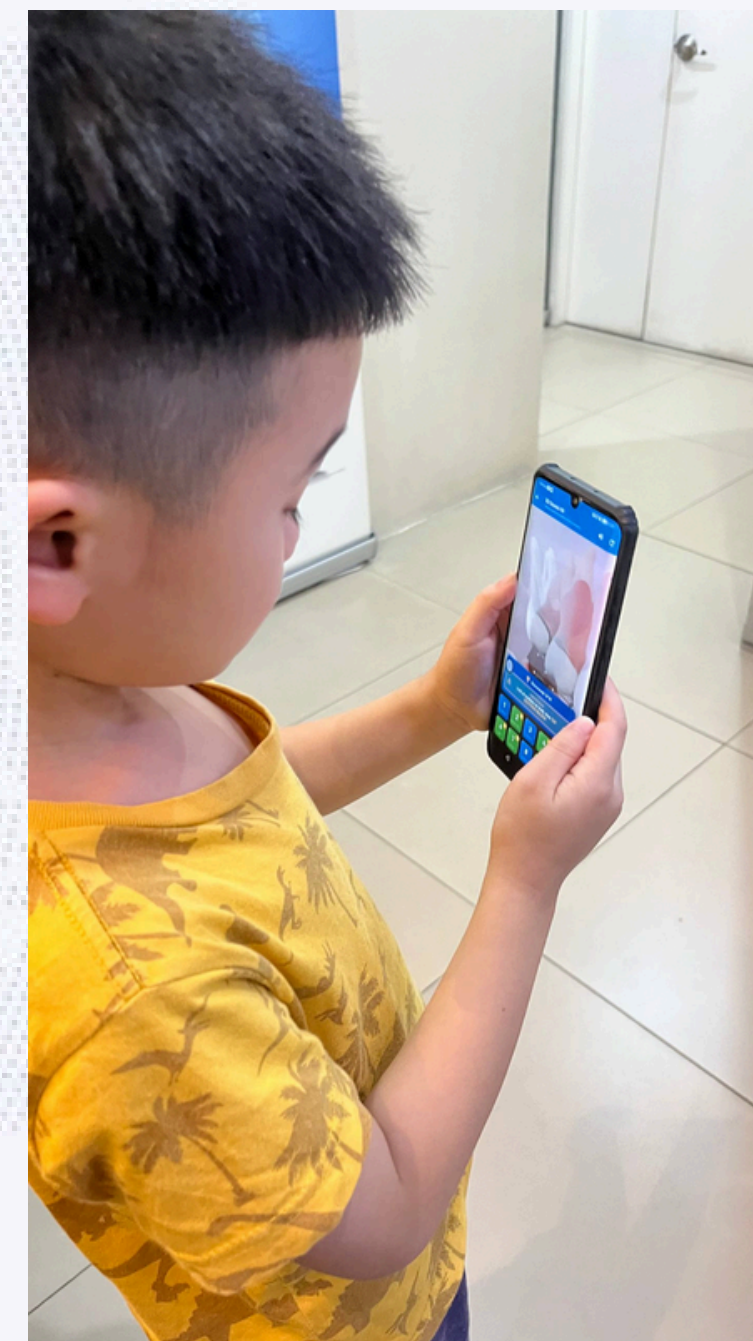
EfficientNetB0 (Improved)
Accuracy: 19.33%

True Label	Calculus	Caries	Healthy_Teeth	Stain	not_teeth
	23	0	0	0	0
	23	0	0	0	0
	23	0	0	0	0
	23	0	0	0	0
	27	0	0	0	0
		Predicted Label			

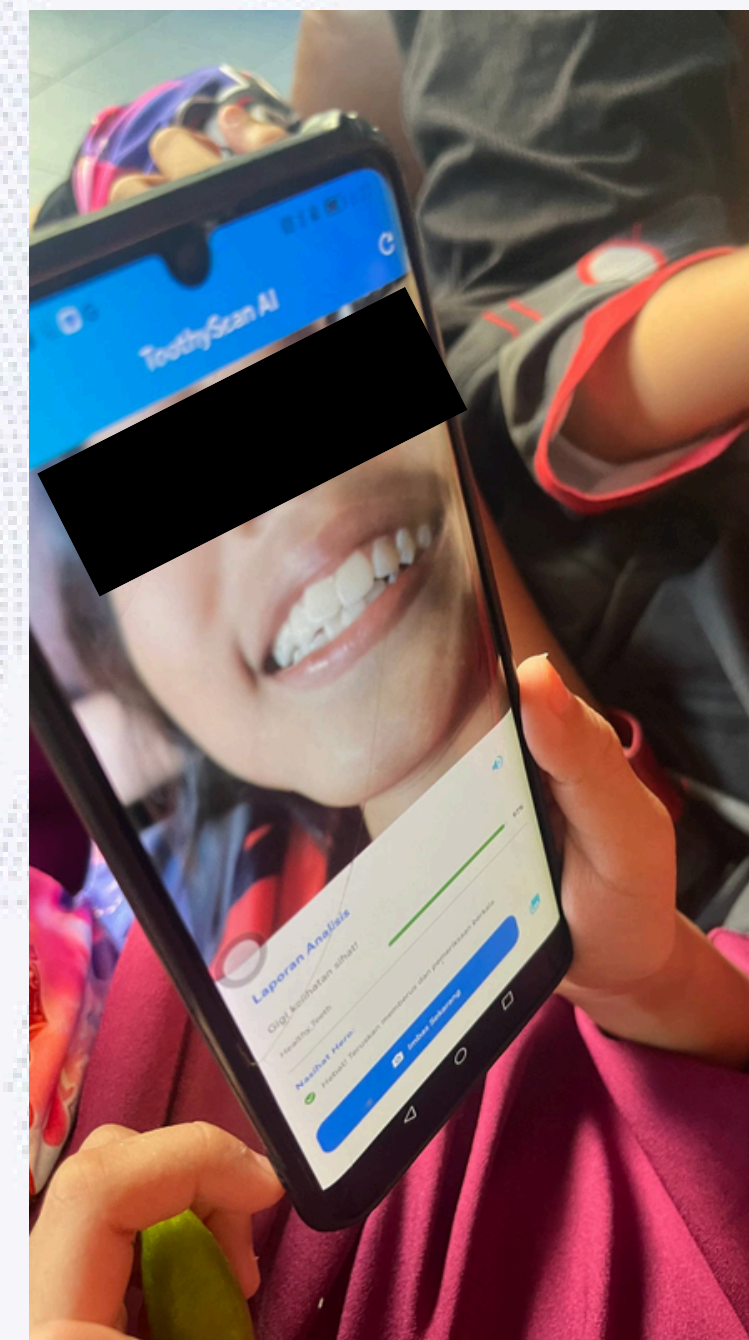
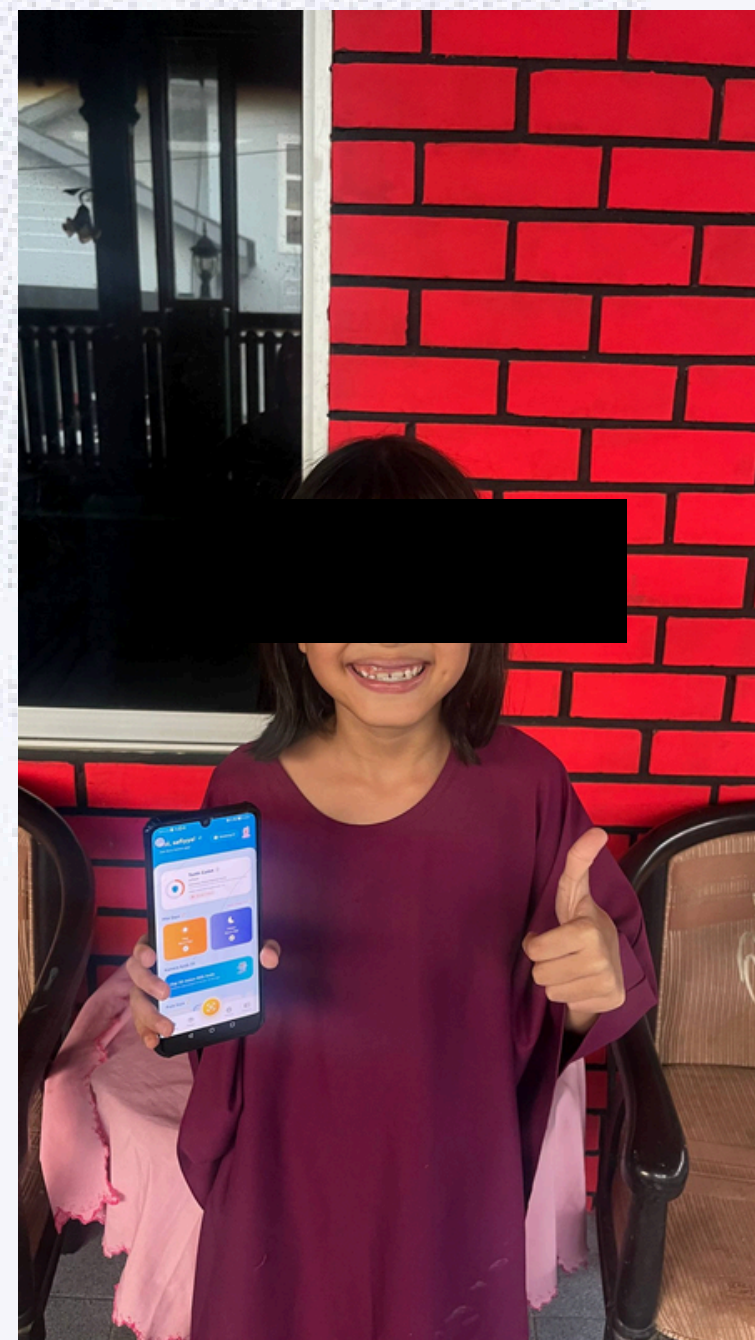
USER TESTING METHODOLOGY (CONT.)



USER TESTING METHODOLOGY (CONT.)



USER TESTING METHODOLOGY (CONT.)



CONCLUSION



Objectives Achieved

- ✓ All project objectives were successfully achieved
- ✓ Core modules were fully implemented:
- ✓ Gamification, E-Learning, AR, AI classification, AI chatbot
- ✓ The system achieved excellent usability and positive user feedback



Future Work & Commercial Potential

- Enhance AI model with multi-label condition detection
- Expand dental condition coverage
- Extend platform support (iOS)
- Deploy as an interactive application in dental clinics for child patients
- Commercial potential as a clinic-based educational tool
- Future extension to adult oral health education



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Demo :)

Thank You